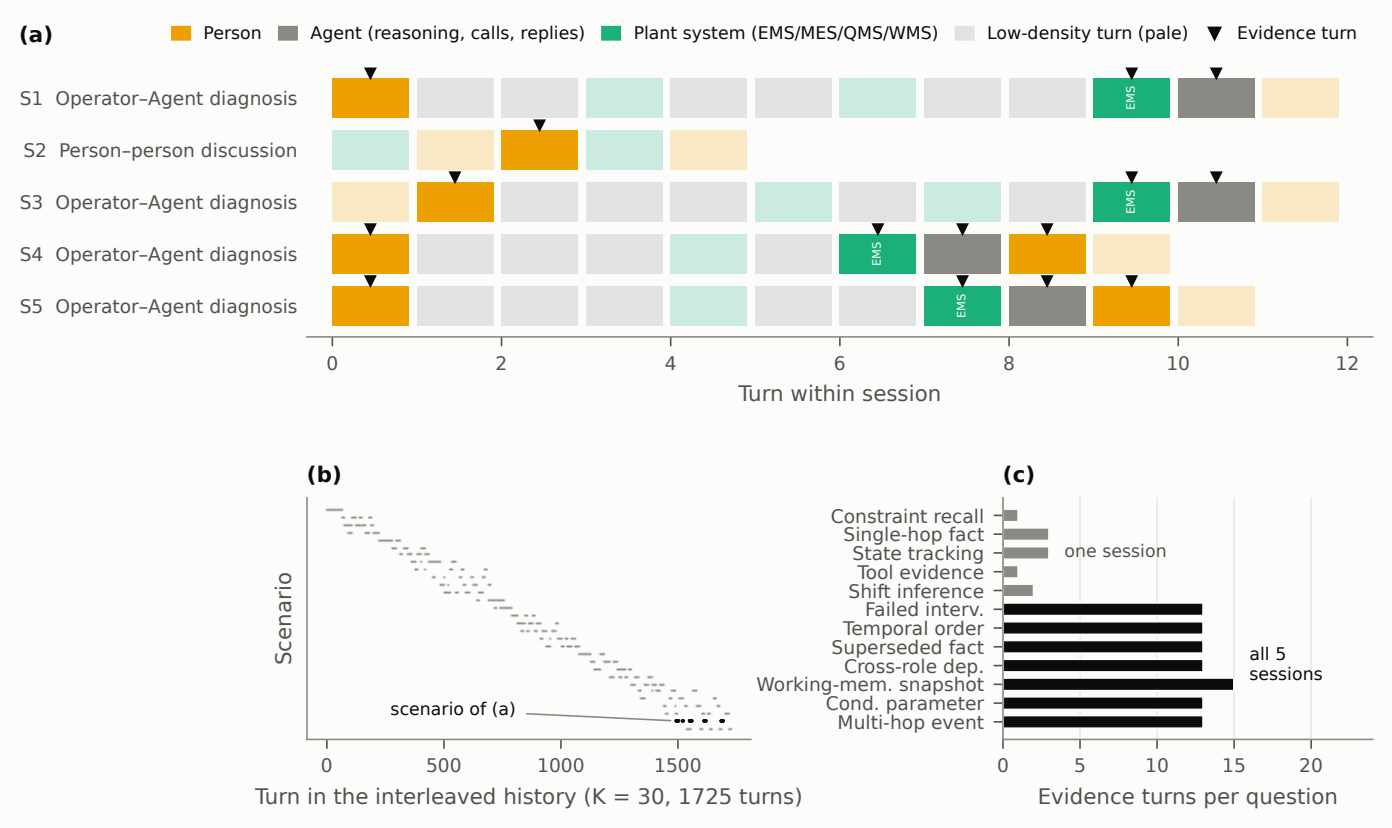
(a)
Person
Agent (reasoning, calls, replies)
Plant system (EMS/MES/QMS/WMS)
Low-density turn (pale)
Evidence turn
S1 Operator–Agent diagnosis
S2 Person–person discussion
S3 Operator–Agent diagnosis
S4 Operator–Agent diagnosis
S5 Operator–Agent diagnosis
EMS
0
2
4
6
8
10
12
Turn within session
(b)
Scenario
scenario of (a)
0
500
1000
1500
Turn in the interleaved history (K = 30, 1725 turns)
(c)
Constraint recall
Single-hop fact
State tracking
Tool evidence
Shift inference
Failed interv.
Temporal order
Superseded fact
Cross-role dep.
Working-mem. snapshot
Cond. parameter
Multi-hop event
one session
all 5 sessions
0
5
10
15
20
Evidence turns per question